

LECTURE#05

MULTIVARIABLE CALCULUS

PARTIAL DERIVATIVE

- Let f be a function of x and y if we hold y constant, say $y = y_0$ and view x as a variable, then $f(x, y_0)$ is function of x -alone.
- If this function is differentiable at $x = x_0$, then the value of this derivative is denoted by $f_x(x_0, y_0)$ and is called the partial derivative of f with respect to x at the point (x_0, y_0) .
- Similarly, If we hold x constant say $x = x_0$ and view y as a variable, then $f(x_0, y)$ is a function of y -alone, if this function is differentiable at $y = y_0$, then the value of the derivative is denoted by $f_y(x_0, y_0)$ and is called the partial derivative of f w. r. t. y at (x_0, y_0) .

EXAMPLE 01

Find $f_x(1, 2)$ and $f_y(1, 2)$ if $f(x, y) = 2x^3y^2 + 2y + 4x$.

Solution: Treating y as a constant and differentiating w. r. t. x , we obtain

$$f_x(x, y) = 6x^2y^2 + 4$$

Treating x as a constant and differentiating w. r. t. y , we obtain

$$f_y(x, y) = 4x^3y + 2$$

Substituting $x = 1$ and $y = 2$ in these partial derivative formulas yield

$$f_x(1, 2) = 6(1)^2(2)^2 + 4 = 28$$

$$f_y(1, 2) = 4(1)^3(2) + 2 = 10$$

EXAMPLE 02

Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ if $z = x^4 \sin(xy^3)$

Solution:

$$\begin{aligned}\frac{\partial z}{\partial x} &= x^4 \cos(xy^3) \cdot y^3 + 4x^3 \sin(xy^3) \\ &= x^4 y^3 \cos(xy^3) + 4x^3 \sin(xy^3)\end{aligned}$$

$$\begin{aligned}\frac{\partial z}{\partial y} &= x^4 \cos(xy^3) \cdot 3xy^2 \\ &= 3x^5 y^2 \cos(xy^3)\end{aligned}$$

EXERCISE

In exercises 1-18 find the first partial derivatives of f .

1. $f(x, y) = 2x^4y^3 - xy^2 + 3y + 1$

2. $f(x, y) = (x^3 - y^2)^2$

3. $f(r, s) = \sqrt{r^2 + s^2}$

4. $f(s, t) = \frac{t}{s} - \frac{s}{t}$

5. $f(x, y) = xe^y + y\sin x$

6. $f(x, y) = e^x \ln xy$

7. $f(t, v) = \ln \sqrt{\frac{t+v}{t-v}}$

8. $f(u, w) = \arctan\left(\frac{u}{w}\right)$

9. $f(x, y) = x \cos\left(\frac{x}{y}\right)$

10. $f(x, y) = \sqrt{4x^2 + y^2} \sec x$

$$11.f(r, s, t) = r^2 e^{2s} \cos t$$

$$12.f(x, y, t) = \frac{(x^2 - t^2)}{1 + \sin 3y}$$

$$13.f(x, y, z) = (y^2 + z^2)^x$$

$$14.f(r, s, v) = (2r + 3s)^{\cos v}$$

$$15.f(x, y, z) = x e^z - y e^x + z e^{-y}$$

$$16.f(r, s, v, p) = r^3 \tan s + \sqrt{s} e^{v^2} - v \cos 2p$$

$$17.f(q, v, w) = \sin^{-1} \sqrt{qv} + \sin vw$$

$$18.f(x, y, z) = x y z e^{xyz}$$

PARTIAL DERIVATIVES OF HIGHER ORDERS

The partial derivative of f_x and f_y of a function f of two variables x and y , may possess derivatives in such cases, the second order partial derivatives are defined as below

$$\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} (f_x) = (f_x)_x = f_{xx}$$

$$\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y} (f_x) = (f_x)_y = f_{xy}$$

CONT...

Similarly,

$$\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} (f_y) = (f_y)_x = f_{yx}$$

$$\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y} (f_y) = (f_y)_y = f_{yy}$$

Thus, there are four second order partial derivatives for a function $z = f(x, y)$.

The partial derivatives of f_{xy} and f_{yx} are called mixed second partials and are not equal the general partial derivatives of order more than two can be defined in a similar manner.

EXAMPLE 01

If $z = \arcsin\left(\frac{x}{y}\right)$ or $\sin^{-1}\left(\frac{x}{y}\right)$ then prove that $\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x}$

$$\frac{\partial z}{\partial x} = \frac{1}{\sqrt{1 - \frac{x^2}{y^2}}} \cdot \frac{1}{y} = \frac{1}{\sqrt{y^2 - x^2}}$$

$$\frac{\partial z}{\partial y} = \frac{1}{\sqrt{1 - \frac{x^2}{y^2}}} \cdot -\frac{x}{y^2} = \frac{-x}{y\sqrt{y^2 - x^2}}$$

CONT...

$$\begin{aligned}\frac{\partial^2 z}{\partial x \partial y} &= \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial y} \right) = \frac{\partial}{\partial x} \left(\frac{-x}{y\sqrt{y^2 - x^2}} \right) \\ &= \frac{-1}{y\sqrt{y^2 - x^2}} - \frac{x}{y} \left(\frac{x}{(y^2 - x^2)^{3/2}} \right) \\ &= \frac{-y^2 + x^2 - x^2}{y(y^2 - x^2)^{3/2}} \\ &= \frac{-y}{(y^2 - x^2)^{3/2}}\end{aligned}$$

CONT...

$$\begin{aligned}\frac{\partial^2 z}{\partial y \partial x} &= \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial x} \right) = \frac{\partial}{\partial y} \left(\frac{1}{\sqrt{y^2 - x^2}} \right) \\ &= \frac{\partial}{\partial y} \left((y^2 - x^2)^{-1/2} \right) \\ &= -\frac{1}{2} (y^2 - x^2)^{-3/2} \cdot 2y \\ &= \frac{-y}{(y^2 - x^2)^{3/2}} = \frac{\partial^2 z}{\partial x \partial y}\end{aligned}$$

EXAMPLE 02

If $f(x, y) = x \cos y + ye^x$ than find $\frac{\partial^2 f}{\partial x \partial y}$, $\frac{\partial^2 f}{\partial y \partial x}$, $\frac{\partial^2 f}{\partial x^2}$, $\frac{\partial^2 f}{\partial y^2}$.

Solution:
$$\frac{\partial f}{\partial x} = \cos y + ye^x$$

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial}{\partial y} (\cos y + ye^x) = -\sin y + e^x$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial}{\partial x} (\cos y + ye^x) = ye^x$$

CONT...

$$\frac{\partial f}{\partial y} = -x \sin y + e^x$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial}{\partial x} (-x \sin y + e^x) = -\sin y + e^x$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial}{\partial y} (-x \sin y + e^x) = -x \cos y$$

EXERCISE

In Exercises 1-13 verify that $w_{xy} = w_{yx}$.

1. $w = xy^4 - 2x^2y^3 + 4x^2 - 3y$

2. $w = x^2/(x + y)$

3. $w = x^3e^{-2y} + y^{-2}\cos x$

4. $w = y^2e^{x^2} + \frac{1}{x^2y^3}$

CONT...

5. $w = \sqrt{x^2 + y^2 + z^2}$

6. If $w = 3x^2y^3z + 2xy^4z^2 - yz$, find w_{xyz} .

7. If $w = u^4vt^2 - 3uv^2t^3$, find w_{tut} .

8. If $u = v \sec(rt)$, find u_{rvr} .

9. If $v = y \ln(x^2 + z^4) - 3uv^2$, find v_{zzy} .

10. If $w = \sin(xyz)$, find $\frac{\partial^3 w}{\partial z \partial y \partial x}$.

CONT...

11. If $w = \frac{x^2}{y^2+z^2}$, find $\frac{\partial^3 w}{\partial z \partial y^2}$.

12. If $w = r^4 s^3 t - 3s^2 e^{rt}$, verify that $w_{rrs} = w_{rsr} = w_{srr}$.

13. If $w = \tan(uv) + 2\ln(u+v)$, verify that $w_{uvv} = w_{vuv} = w_{vvu}$.